

## 1. Multi-Year Sales Performance Analysis (Grouped Bar Chart)

Using a dataset containing yearly sales data for multiple products, visualize the sales performance trends using a **grouped bar chart**.

**Columns:** Year, Product, Sales

**Requirements:**

- Display sales comparison between products for each year.
- Add appropriate axis labels, a legend, and a meaningful title.
- Customize colors and spacing for clarity.
- Interpret which product shows the highest growth trend.

## 2. Climate Change Visualization (Multi-Line Chart)

Create a **multi-line chart** showing temperature trends for three cities over the last 10 years.

**Columns:** Year, City, Average\_Temperature

**Requirements:**

- Use distinct colors and line styles for each city.
- Add gridlines and annotations for record-high or record-low years.
- Include a clear title such as *“Decade-Long Temperature Trends Across Cities”*.
- Analyze which city experienced the steepest temperature rise.

## 3. Energy Consumption Composition (Pie + Donut Chart)

Visualize how different energy sources contribute to total consumption using a **donut chart** (pie chart with a hole).

**Columns:** Energy\_Source, Percentage

**Requirements:**

- Show labels and percentages on the chart.
- Highlight renewable sources with distinct colors.
- Add an annotation summarizing renewable vs. non-renewable proportions.
- Interpret what this distribution reveals about sustainability.

## 4. Employee Salary Analysis (Box Plot + Histogram)

Analyze the distribution of employee salaries across departments using **box plots** and a **histogram**.

**Columns:** Department, Salary

**Requirements:**

- Create side-by-side box plots for each department.

- Use a histogram to visualize overall salary distribution.
- Detect outliers and describe their potential causes.
- Add the median, mean, and interquartile range (IQR) annotations.

## 5. Study Efficiency Correlation (Scatter Plot + Regression Line)

Explore the correlation between study hours and exam performance using a **scatter plot** with a regression line.

**Columns:** Study\_Hours, Exam\_Score

**Requirements:**

- Plot individual data points and fit a linear regression line.
- Display the equation and  $R^2$  value on the chart.
- Use color intensity to indicate study consistency (optional 3rd variable)